

S.No. : 632

NCS 4201

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Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 43202

Roll
No.

1	1	3	0	4	3	9	3	6	0
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B. Tech. Examination 2023-24

(Even Semester)

PROGRAMMING CONCEPTS WITH PYTHON

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION - A

1. Attempt all parts of the following : $8 \times 1 = 8$

(a) Write the code to create a dictionary of three fruits.

(b) What do you mean by len () function?

(c) What is doc string?

(d) What are tuples in Python?

[P. T. O.]

- (c) What type of error is returned by the following code :

```
S = "python practice"
```

```
S[5] = d
```

- (f) A tuple is declared as `list = (1, 2, 3, 4, 5)`. What will be the value of `print(list[: :-1])`?
- (g) What is module and package in Python?
- (h) Name any two files access modes in Python.

SECTION - B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) What is meant by selection, iteration and sequence control structures?
- (b) Write the algorithm, pseudocode and draw the flowchart for the following :
- (i) Find minimum in a list.
 - (ii) Guess an integer number in a range.
- (c) What is a function? Mention the type of function and use.
- (d) Write a Python program to check if a 3 digit number is Armstrong number or not?

SECTION - C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

3. (a) Describe the syntax for the following function and explain with an example :
- (i) `abs()`
 - (ii) `max()`
 - (iii) `divmod()`
 - (iv) `pow()`
 - (v) `len()`
- (b) Write Python program to implement stack operation.
- (c) Write Python program to count the number of times an item appears in the list.
4. (a) Explain identifier, keywords, statements, expressions and variables in Python programming language with examples.
- (b) Explain the use of `join()` and `split()` string methods with examples. Describe why strings are immutable with an example?

[P. T. O.]

- (c) Describe the characteristics of the CSV format.
5. (a) Explain the concept of scope and life time of variables in Python programming language with an example.
- (b) Write Python program to check for the presence of a key in the dictionary and find the sum of all its values.
- (c) Discuss the following dictionary methods with an example :
- (i) get ()
 - (ii) pop ()
 - (iii) values ()
 - (iv) items ()
6. (a) Write Python program to change the file extension from .txt to .csv of all the files (including from sub directories) for a given path.
- (b) Discuss format specifiers and escape sequence with examples.

- (c) Discuss the following methods of OS.path module :
- (i) isfile ()
 - (ii) abspath ()
 - (iii) dirname ()
 - (vi) relpath ()

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